Answer key: Lesson 8, Balance and Swap

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill in the missing numbers.

Answer: 4, 5, 3, 2

(a) 5 + 3 is 8 and 4 + 4 is 8. (b) 6 + 4 is 10 and 5 + 5 is 10. (c) 7 + 3 is 10 and 3 + 7 is 10. (d) 2 + 8 is 10 and 8 + 2 is 10.

Watch for: Writing the total of the left side in the blank (8 in part a).; Thinking the equals sign means the answer comes next.

S2. Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

Answer: 1 more button (6 + 2 = 8 and 7 + 1 = 8)

The left plate has 6 + 2 = 8 buttons. The right plate has 7 buttons, so it needs 1 more to make 8.

Watch for: Drawing 8 more buttons on the right plate.; Forgetting the 7 buttons already there.

S3. Siti writes 6 + 3 = 5 + 9.

Answer: No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next.

6 + 3 is 9. The right side must also be 9. 5 and 4 more make 9, so the blank is 4. Siti wrote 9, the total of the left side, but the blank is only one of the two numbers on the right.

Watch for: Saying yes because 9 is the answer to 6 + 3.; Writing 9 in the box again (copying Siti).

S4. 7 + 5 = 12. Fill in the missing numbers.

Answer: 6, 7, 8

(a) 6 + 6 = 12. (b) 5 + 7 = 12. (c) 4 + 8 = 12. The first number goes down by 1 and the second goes up by 1, so the total stays 12.

Watch for: Leaving the second number as 5.; Making the second number go down as well.

S5. Fill in the missing numbers.

Answer: 10, 10, 30, 40

(a) 20 went up by 20 to 40, so 30 goes down by 20 to 10. (b) 50 went up by 10 to 60, so 20 goes down by 10 to 10. (c) 40 went down by 10 to 30, so 20 goes up by 10 to 30. (d) 30 went down by 10 to 20, so 30 goes up by 10 to 40.

Watch for: Writing the total of the left side (50, 70, 60 or 60).; Moving both numbers the same way (part c: 20 - 10 = 10).

S6. Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?

Answer: 5 blue pens (9 + 6 = 10 + 5)

Mei has 9 + 6 = 15 pens. Ben has 10 red pens, so he needs 5 more to make 15. Another way: his red pens are 1 more than Mei's, so his blue pens are 1 fewer: 5.

Watch for: Answering 6 (copying Mei's blue pens).; Answering 15 (the total).

S7. Fill in the missing number.

Answer: 45 (35 + 12 = 47 and 45 + 2 = 47)

The 12 went down by 10 to become 2, so the 35 goes up by 10 to become 45. Check: 35 + 12 = 47 and 45 + 2 = 47.

Watch for: Writing 47 (the total).; Writing 25 (taking a ten from the 35).

Think harder (page 3)

T1. Mei says 47 + 18 and 45 + 20 have the same total. Ali says 45 + 20 has the greater total. Do not add.

Answer: Mei. 47 went down by 2 to 45 and 18 went up by 2 to 20. The 2 moved across, so both totals are 65.

47 to 45 is 2 less. 18 to 20 is 2 more. The 2 moved from one number to the other, so the total stays the same. Both sums make 65.

Watch for: Saying 45 + 20 is bigger because 20 is a bigger number than 18.; Adding both sums and not saying how the numbers moved (correct, but check the reason).

T2. Do not add. Circle LEFT or RIGHT for the greater total, or circle SAME.

Answer: SAME, SAME, RIGHT

(a) 29 up 1 to 30, 14 down 1 to 13: SAME (both 43). (b) 53 down 3 to 50, 20 up 3 to 23: SAME (both 73). (c) 26 up 2 to 28 and 15 up 1 to 16: both went up, so RIGHT is bigger (41 and 44).

Watch for: Saying SAME for part c because both numbers changed.; Looking only at the biggest numbers.

T3. Fill in the missing number. Then find the total.

Answer: 25, then the total is 75

49 went up by 1 to 50, so 26 goes down by 1 to 25. 50 + 25 = 75, so 49 + 26 = 75.

Watch for: Writing 27 (making 26 go up as well).; Writing 75 in the blank.

T4. Ben writes 24 + 17 = 26 + 19.

Answer: No. 24 + 17 = 41 but 26 + 19 = 45. Both numbers went up by 2. Change 19 to 15.

24 went up by 2 to 26, so 17 should go down by 2 to 15. Ben made it go up by 2 to 19, so the right side is 4 too big. Check: 24 + 17 = 41 and 26 + 15 = 41.

Watch for: Saying it is balanced because both numbers changed by 2.; Changing the 26 and not saying which number you change.

Go further (page 4)

G1. Fill in the missing numbers so that all three sums have the same total.

Answer: 27 and 17 (every sum is 47)

19 up 1 to 20, so 28 down 1 to 27. The total is 20 + 27 = 47 (check: 19 + 28 = 47). Then [] + 30 = 47, so the second blank is 47 - 30 = 17.

Watch for: Writing 27 in both blanks.; Writing 30 or 28 as the second blank.

G2. Fill in the blanks so both sides are the same. Write 3 different pairs. Many answers are right.

Answer: Any pair where the first blank is 4 more than the second, for example 5 and 1, 6 and 2, 7 and 3 (also 4 and 0).

If the second blank is 1, the right side is 10, so the first blank is 5. If it is 2, the right side is 11, so the first blank is 6. The first blank is always 4 more than the second, because 9 is 4 more than 5.

Watch for: Writing the same number in both blanks.; Giving one pair only.

G3. Mei has 28 stickers on a page and 17 in her pocket. Ali, Ben and Kumar have the same total as Mei. Fill in the blanks.

Answer: 15, 20 and 33 (Mei has 45 stickers; 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45)

Mei has 28 + 17 = 45. Ali: 30 is 2 more than 28, so his pocket has 2 fewer than 17: 15. Ben: 25 is 3 fewer than 28, so his pocket has 3 more than 17: 20. Kumar: his pocket 12 is 5 fewer than 17, so his page has 5 more than 28: 33. Check: 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45.

Watch for: Writing 17 for Ali's pocket.; Making Ben's pocket 14 (moving the wrong way).; Writing 17 or 45 for Kumar's page.

Investigation (page 5): Keep it the same

First step: 7 + 5 is not 10 (the totals are 10, 10 and 12). From 6 + 4 to 7 + 3 the 4 goes down 1. Next: add some to one number and take the same from the other and the total stays the same (give and take). Add 3 to both: the total goes up by 6. Take 2 from both: the total goes down by 4. When both numbers change the same way, the total changes by double.

Look for: A child who tries more than one pair of numbers, says it stays the same when one goes up and the other goes down by the same amount, and notices that changing both numbers the same way changes the total by double. Any clear reason is fine.

Exit ticket (page 5)

E1. Fill in the missing number.

Answer: 4

9 went up by 1 to 10, so 5 goes down by 1 to 4. Both sides make 14.

Watch for: Writing 14 (the total).; Writing 6 (making the 5 go up).

E2. Is 28 + 6 = 30 + 4 balanced?

Answer: Yes

28 went up by 2 to 30 and 6 went down by 2 to 4. Both sides make 34.

Watch for: Writing no because the numbers look different.

E3. Fill in the missing number.

Answer: 2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2.

30 went up by 10, so 12 goes down by 10 to 2. Both sides make 42.

Watch for: Writing 42 (the total).; Writing 22 (making the 12 go up).

Sneak-peek puzzles (not in the workbook)

1. Yes, the same: 10 beads each.
2. 4. 9 is 1 more than 8, so the blank is 1 less than 5. (Both sides make 13.)